

JHU WebSec Application Container Security Module 5 Assignment

V.25.2

Objectives

1. Demonstrate proficiency working with images, containers, and container repositories. Implement Docker. Provide annotated screen snips in a Word document.
2. Demonstrate proficiency working with security solutions for containers – Implement Anchore Grype.
3. Generalize the security concerns regarding containers. Scan an image, conduct a threat model, research vulnerabilities, and provide a write-up.

Introduction

This week, we will look at containers and then use a security scanning tool called Anchore Grype. We will use Docker to run an image in a container. get you some hands-on experience with Docker and an appreciation for it; although containers are simple to use, their security cannot be neglected.

Recommend using a new Virtual Machine sandbox. Set the memory in your VMs to at least 4GB!

For reference: [GitHub - anchore/grype: A vulnerability scanner for container images and filesystems](#)

Part 1: Installing Docker

Follow the instructions located here: <https://docs.docker.com/engine/install/ubuntu/>

Optional: Run Docker without sudo

Instead of using **sudo** for every Docker command, add the user to the **docker** group

```
$ sudo usermod -a -G docker student
```

Exit out of your current shell (or even reboot) and log back in.

If you run this step for the rest of the assignment, you won't need **sudo when running docker**.

Deliverable PART 1: Create a Word document and add a section called: "Part 1: Installing Docker". Please provide a screenshot of the output from the command: `$sudo docker run hello-world`.

Part 2: OWASP Juice Shop

1. Load the OWASP juice shop:

```
$ sudo docker pull bkimminich/juice-shop
```

2. Run the Juice Shop website

```
$ sudo docker run --rm -p 3000:3000 bkimminich/juice-shop
```

3. Open a browser and navigate to **http://localhost:3000/**

Deliverable PART 2: Add to the Word doc a section called: “Part 2: Running OWASP Juice Shop”. Please provide a screenshot of the output from the Juice Shop.

Part 3: Installing Anchore Grype

1. Navigate to <https://github.com/anchore/grype/releases>
2. Download the lasted version. Ex: grype_0.80.1_linux_amd64.tar.gz
3. Extract grype

```
$ tar -xvzf grype_0.62.2_linux_amd64.tar.gz
```

4. Confirm version:

```
$ ./grype version
```

Deliverable PART 3: Add a section to the Word doc called “Part 3: Installing Anchore Grype.” Please provide a screenshot showing the version and successful installation of Anchore Grype.

Part 4: Scanning Images with Anchore Grype

1. Scan Ubuntu Image

```
$ ./grype ubuntu:latest | grep 'Medium\|Critical' | more
```

2. Scan Juice Shop image

```
$ ./grype bkimminich/juice-shop | grep 'Medium\|Critical' | more
```

3. Scan local file system

```
$ ./grype dir:/ | grep 'Medium\|Critical' | more
```

Deliverable PART 4: Add a section to the Word doc, add a section called “PART 4: Scanning Images”. Provide annotated screen snips of the successful completion of each of the above scans.

Part 5: Research vulnerabilities and provide analysis

1. Select two of the vulnerabilities (CVE) found in your scan results.
2. Look them up in:
 - a. CVE - Search CVE List (mitre.org) [CVE - Search CVE List](#)
 - b. [Known Exploited Vulnerabilities Catalog | CISA](#)
 - c. Or other repository

Deliverables PART 5: Add a section to the Word doc, “PART 5: Vulnerability Analysis”

1. List the vulnerabilities and briefly describe them.
2. Provide a paragraph with a brief analysis of the results using the Grype scanner.
3. Would or would you not recommend using any of these images in production?
Please provide your rationale.

Grading Rubric

Objective	Value
1. Demonstrate proficiency working with image, a container, and a container repository Part 1: Implement Docker, provide annotated screen snip in a Word doc.	10
2. Demonstrate proficiency in running a Docker container. Part 2: Running OWASP Juice Shop with screen snip	20
3. Demonstrate proficiency running Anchore Grype image security scanner. Part 3: Annotated screen snip showing installation of Anchore Grype	20
4. Demonstrate proficiency in scanning images with Grype security scanner.	20

Part 4: Annotated screen snips of the above scans.	
5. Generalize the security concerns regarding container images: Part 5: research 2 vulnerabilities, provide a write-up demonstrating applied knowledge of using a container security scanner.	20
Does the submission look professional, easy to read, clean presentation, with spelling and grammar at master's level?	10
TOTAL	100